

4(a)	<i>Max 3 for suberin, max 3 for lignin</i> any four from: 1 lignin / suberin, is, hydrophobic / (partially) impermeable to water / AW ; e.g. waterproof / barrier to water <i>suberin</i> 2 suberin found in, Casparian strip / endodermal cells / endodermis ; 3 water must move from apoplast (pathway) to, symplast (pathway) / cytoplasm (at endodermis) ; 4 allows control of solutes / AW, before entering, endodermal cells / xylem ; A helps to prevent, toxins / bacteria / AW, entering xylem <i>in context of mp3</i> <i>lignin</i> 5 lignin found in, cell walls of xylem vessel elements / walls of xylem vessels ; A xylem / vessel, elements 6 <i>idea of</i> lignin / lignified wall, prevents water leaving xylem / loss of water from xylem ; 7 lignified walls, prevents, (inward) collapse of / compression of / AW, xylem ; R bursting 8 provides strength / secondary thickening, for (structural) support (of xylem) ; 9 AVP ; e.g. <i>ref. to</i> lignin as part of endodermis adhesion of water molecules to hydrophilic parts of lignin prevents collapse of water column / AW lignification causes death of xylem vessel elements <i>context of producing empty xylem vessels for transport</i>	4
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Question	Answer	Marks
4(b)	<i>max 4 if no reference to laccase / monolignols</i> <i>allow substrate(s) for monolignols</i> any five from: 1 laccase uses induced fit ; 2 <i>idea that</i> active site shape not fully complementary to (shape of) monolignols ; 3 active site changes shape to, fit / bind to, monolignols ; AW 4 becomes fully <u>complementary</u> ; 5 forming enzyme-substrate complex ; 6 lowers, activation energy / energy of activation; 7 suggestion of how activation energy is lowered ; 8 laccase returns to original shape, after product leaves active site / for re-use / AW ; 9 AVP ; e.g. <i>suggestion that</i>, copper ions are, cofactors / necessary for enzyme action <i>suggestion that</i> monolignols are held in place by (temporary) H bonds	5